A Treatise on Differential Equations of the Ordinary Kind

With Notes on Integration

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Contents

0 Another Try at Nets

Uh yeah . . . nets. Basically what I have been trying to get to click the whole past quarter. Attempt 2 . . .

0.1 First Countability

What has been giving us sequential equivalence in $\mathbb{R}$ is the first countability of $\mathbb{R}$ (which means around any point x, there exists a countable basis). We used this to prove the more general properties implied sequential properties (by using the first countable property to construct/apply subsequences).

All metric spaces are first countable; but there are some topological spaces that are not first countable. Consider the following topology on $\mathbb{R}$:

$$\tau := \{\varnothing\} \cup \{\mathbb{R} \setminus F : F \text{ is finite}\}$$

Now take some point $x \in \mathbb{R}$ and assume it has some countable basis $\mathcal{B}_x := \{B_n\}_{n\in\mathbb{N}}$, in which case:

$$\forall n \in \mathbb{N} : B_n = \mathbb{R} \setminus F_n$$

Then, the following set should be countably infinite:

$$\bigcup_{n\in\mathbb{N}} (\mathbb{R} \setminus B_n)$$

since each F_n is finite. We note that:

$$\bigcup_{n\in\mathbb{N}} (\mathbb{R} \setminus B_n) = \mathbb{R} \setminus \bigcap_{n\in\mathbb{N}} B_n = \mathbb{R} \setminus \{x\}$$

which is a contradiction, since $\mathbb{R}$ is not countably infinite and removing one point does not change this fact. Notice the intersection of the basis elements is solely x. That is because the topology is T_1, which means we can find some open set containing x that separates x and any other arbitrary point. Then by basis definition, there is some $B_n \subseteq O_x$.

This same T_1 condition idea with basis elements gives equivalence of a limit point and ω-limit point in the next section.

In such spaces; if the sequential condition holds, then the general condition holds. E.g.:

1. Sequentially Compact $\implies$ Limit Point Compact
2. Sequentially Closed $\implies$ Topologically Closed

but the same does not hold for the converse.

0.2 Nets . . . Finally

We begin by introducing the notion of a directed set:

Definition 0.1. *We define an ordering relation $\leq$ for the set D and call it a directed set under the following conditions:*

1. *Reflexive:* $\forall x \in D : x \leq x$

2. *Transitive:* $\forall x, y, z \in D : x \leq y \wedge y \leq z \implies x \leq z$
3. *Directed:* $\forall x, y \in D, \exists z \in D : (x \leq z \wedge y \leq z) \vee (z \leq x \wedge z \leq y)$

Notice that the first two properties align with a partial order. However a partial order also requires anti-symmetry, which means:

$$x \leq y \wedge y \leq x \implies x = y$$

which is not required for a directed set. This will become quite important later on. The third property (which is direction) tells us whether the net is upward directed (the left side condition) or downward directed (the right side), or both! We view an equivalent property, for which WLOG, we assume an upward directed set D:

Lemma 0.1. *Given $(D, \leq)$ an upward directed set, then for all finite subsets S of D:*

$$\forall S \subseteq D, \exists z \in D, \forall x \in S : x \leq z \iff \forall x, y \in D, \exists z \in D : x \leq z \wedge y \leq z$$

In other words, all finite subsets of the directed set admit an upper bound.

Proof. $\implies$: The forward direction is trivial as a two element subset is finite.

$\impliedby$: For the converse we prove via induction. The base case is a finite set of two elements, which we are given is true. Now assume it holds for a finite set of n elements such that:

$$S_n \coloneqq \{x_i : 0 \leqslant i \leqslant n-1\} \wedge \forall x_i \in S_n, \exists z \in D : x_i \leq z$$

Now consider $S_{n+1} \coloneqq S_n \cup \{x_n\}$ where x_n is different from all elements in S_n. Then we get that, by the base case:

$$\exists w \in D : x_n \leq w \wedge z \leq w$$

and finally by transitivity, we obtain that $\forall x_i \in S_{n+1} : x_i \leq w$. $\square$

For the remainder of these notes, we assume a directed set D is upward directed unless stated otherwise (the downward directed set notion will also be useful). Now a net is built on a directed set by the following:

Definition 0.2. *A net is a function $\omega : D \to X$ where D is a directed set.*

We define net convergence as the following. Now, to bring about the similarity to sequences, we shall denote a net as $\{x_\alpha\}_{\alpha \in D}$ where D is the directed set. This is still a function where $x_\alpha = \omega(\alpha)$, but this representation shows D as an indexing set.

Now in full generality of an arbitrary topological space, we say that the net converges or that $x_\alpha \to x$ if:

$$\forall U_x \in \tau_X, \exists \alpha_0 \in D, \forall \alpha \geq \alpha_0 : x_\alpha \in U_x$$

where U_x is an arbitrary open neighborhood of the point $x \in X$ with the topology τ_X. For the majority of these notes, we shall work in metric spaces (X, ϱ) (unless stated otherwise), so we can instead say:

$$\forall \epsilon > 0, \exists \alpha_0 \in D, \forall \alpha \geq \alpha_0 : x_\alpha \in B(c, \epsilon) \iff \varrho(x_\alpha, x) < \epsilon$$

so if our directed set $D = \mathbb{N}$, we obtain a sequence. We can alternatively characterize convergence by the following definition:

Definition 0.3. *The tail of a net at α_0 is defined as:*

$$T_{\alpha_0} \coloneqq \{x_\alpha : \alpha \geq \alpha_0\}$$

Then a net converges to some x if and only if $\forall \epsilon > 0, \exists \alpha_0 \in D$ such that:

$$T_{\alpha_0} \subseteq B(x, \epsilon)$$

We say that a net converges if it is eventually in all open neighborhoods of x. We introduce a related concept from the lens of sequences:

Definition 0.4. *A sequence frequently visits a set A if:*

$$\forall n_0 \geqslant 0, \exists n \geqslant n_0 : x_n \in A$$

So a cluster point of a sequence is defined as the following:

Definition 0.5. *x is a cluster point of $\{x_n : n \in \mathbb{N}\}$ if $\{x_n\}_{n\in\mathbb{N}}$ frequently visits all open neighborhoods of x.*

The notion of a cluster point is something we will bring about when talking about nets and sequences. Considering metric spaces, we get the definition of a cluster point as:

$$\forall \epsilon > 0, \forall n_0 \geqslant 0, \exists n \geqslant n_0 : x_n \in B(x, \epsilon)$$

which permits $x_n = x$ unlike a **limit point of a set**.

1 Compactness

We will now discuss the equivalence of topological and sequential compactness in the reals and in general metric spaces. Let us view the following theorem:

1.1 Limit Points

We start off with a topological proof, which will be key to discussing limit and adherent points:

Lemma 1.1. *The closure of a set A is equal to all of its adherent points.*

Proof. $\implies$: We prove by contrapositive. If x is not an adherent point, then:

$$\exists \epsilon > 0 : B(x, \epsilon) \cap A = \varnothing \implies B(x, \epsilon) \subseteq X \setminus A$$

So x is an interior point of $X \setminus A$, which means:

$$x \in \operatorname{int}(X \setminus A) = X \setminus \overline{A}$$

$\impliedby$: We once again prove by contrapositive. If $x \notin \overline{A}$, then:

$$x \in X \setminus \overline{A} \subseteq X \setminus A \implies x \in X \setminus A$$

Since $x \setminus \overline{A}$ is open, then x must be an interior point, so:

$$\exists \epsilon > 0 : B(x, r) \subseteq X \setminus A \implies \exists \epsilon > 0 : B(x, r) \cap A = \varnothing$$

□

The limits of sequences ARE NOT necessarily limit points. Given some subset $E \subseteq \mathbb{R}$, x is a limit point of E if:

$$\forall \epsilon > 0 : B(x, \epsilon) \cap E \neq \varnothing$$

Notice how this only requires the intersection to be nonempty; which means there could possibly be a finite number of points in the intersection (but we will see this is not possible). Let us look at the following lemma:

Lemma 1.2. *x is a limit point of E if and only if*

$$\exists \{x_n\}_{n \in \mathbb{N}} \in (E \setminus \{x\})^{\mathbb{N}} : x_n \to x$$

Proof. $\implies$: If x is a limit point, then using the first countability of metric spaces, consider:

$$\epsilon_n \coloneqq \frac{1}{n+1}$$

Then by axiom of choice, choose $x_n \in B(x, \epsilon_n) \cap E \setminus \{x\}$, so that:

$$\forall n \in \mathbb{N} : |x_n - x| < \frac{1}{n+1} \implies x_n \to x$$

$\impliedby$: Now for the converse. If there exists such a sequence $\{x_n\}_{n \in \mathbb{N}}$ such that:

$$\forall \epsilon > 0, \exists n_0 \geqslant 0, \forall n \geqslant n_0 : |x_n - x| < \epsilon$$

then we see that

$$\forall \epsilon > 0 : B(x, \epsilon) \cap E \setminus \{x\} \neq \varnothing$$

as $\forall n \geqslant n_0 : x_n \in B(x, \epsilon)$ and by definition of the sequence $x_n \in E \setminus \{x\}$. Thus proving equivalence. □

Now we introduce the following notion:

Definition 1.1. *An ω-limit point of a set E is one for which there exists a countably infinite number of points of E within each punctured ϵ-ball around x.*

It follows that every limit point in $\mathbb{R}$ is an ω-limit point from Lemma 1.2, the sequential definition. This is because:

$$\forall \epsilon > 0, \exists n_0 \geqslant 0, \forall n \geqslant n_0 : x_n \in B(x, \epsilon) \cap E \setminus \{x\}$$

and the natural numbers n_0 and beyond are countably infinite (an injection of $\mathbb{N}$ into this set can easily be shown), thus x is an ω-accumulation point. The weakest sufficient condition is a T_1 space for such equivalence.

Now if the set E consists of some sequence $\{x_n\}_{n\in\mathbb{N}}$, then $E \coloneqq \{x_n : n \in \mathbb{N}\}$, we see that x is a limit point of E if and only if there exists a convergent **subsequence** converging to x. This is not immediately trivial as one of the conditions we require for a subsequence is that:

$$\forall \{n_k\}_{k\in\mathbb{N}} : n_k < n_{k+1s}$$

Observe the definition of sequential closure of some subset E:

Definition 1.2. *$\overline{E}$ is the closure of E if and only if*

$$\forall \{x_n\}_{n\in\mathbb{N}} \in E^{\mathbb{N}} : x_n \to x \implies x \in \overline{E}$$

So by the equivalence stated in Lemma 1.2, $\{x_n\}_{n\in\mathbb{N}}$ belongs to $E^{\mathbb{N}}$ as well (it is not guaranteed that $x \in E$), but by sequential closure x or the limit point must belong in $\overline{E}$.

Now steering our attention back to sequences, the limit of a sequence may not be a limit point. Consider an eventually constant or constant sequence. We see that it converges to whatever value it is eventually constant at; but there is an empty ϵ-punctured ball intersection. So a limit point is a limit of a specific type of sequence, but not vice versa.

This distinction will be quite crucial in the following section.

1.2 Sequential Compactness

Theorem 1.1 (Sequential Bolzano Weierstrass)**.** *Any bounded sequence has a convergent subsequence.*

To prove this fact, we show first that any sequence in $\mathbb{R}$ has a monotonic subsequence.

Proof. Let us define the following set, we call "peaks" of a sequence:

$$P \coloneqq \{n \in \mathbb{N} : (\forall j > n : x_j < x_n)\}$$

In the case that P is countably infinite, then we set:

$$n_0 \coloneqq \inf P \wedge \forall k > 0 : n_k \coloneqq \inf \left(P \setminus \{n_0, ..., n_{k-1}\}\right)$$

In which case we are guaranteed that $\forall k \in \mathbb{N} : n_{k+1} > n_k$ which tells us that $x_{n_{k+1}} < x_{n_k}$, showing that there exists a strictly decreasing subsequence. In the case that P is finite, we choose the following:

$$n_0 \coloneqq \sup(P) + 1$$

that way the peaks are fully ommited from the subsequence, as they will prove to be problematic in the construction. Then for any $k \in \mathbb{N}$, we are guaranteed that $n_k \notin P$ (since $\{n_k\}_{k\in\mathbb{N}}$ is strictly increasing and $n_0 \notin P$), which means:

$$\exists j > n_k : x_j \geqslant x_{n_k}$$

So we simple define:

$$\forall k \in \mathbb{N} : n_{k+1} \coloneqq \inf\{j > n_k : x_j \geqslant x_{n_k}\}$$

in which case we obtain a non-decreasing subsequence. Now any bounded sequence has a monotone subsequence, which converges. What this means is given some sequence $\{x_n\}_{n\in\mathbb{N}}$, we have that:

$$\exists y \in \mathbb{R}, \exists r > 0, \forall n \in \mathbb{N} : x_n \in B(y, r)$$

so the sequence has a convergent subsequence within the ball, but such a subsequence may not converge within the ball. This is due to the topological idea that the limit point certainly belongs in the closure of the set but not necessarily in the open set. Consider the bounded sequence:

$$\{x_n\}_{n\in\mathbb{N}} \in (0,1)^{\mathbb{N}} : \left(\forall n \in \mathbb{N} : x_n \coloneqq \frac{1}{n+2}\right)$$

Then $x_n \to 0$ yet $0 \notin (0, 1)$. This idea of closedness plays a key role in sequential compactness. □

Definition 1.3. *A sequentially compact interval of $\mathbb{R}$ is one for which every sequence has a convergent subsequence that converges to a point within the interval.*

The previous Theorem 1.1 leads to a key theorem:

Theorem 1.2. *An interval of $\mathbb{R}$ is sequentially compact if and only if it is closed and bounded.*

$\Longleftarrow$: This direction follows from Sequential Bolzano Weierstrass. If the interval is bounded, then any sequence has a convergent monotone subsequence. Since it is closed, the limit of the sequence, which is an adherent point, belongs in the closure.

$\Longrightarrow$: Consider any convergent sequence $\{x_n\}_{n\in\mathbb{N}}$ such that $\forall n \in \mathbb{N} : x_n \in I$. Then by sequential compactness:

$$\exists \{n_k\}_{k\in\mathbb{N}} : x_{n_k} \to x \wedge x \in I$$

which forces $x_n \to x$. Thus all convergent sequences converge to points within the set (which by sequential closure) means I is closed. Now assume the interval is not bounded, in which case:

$$\forall r > 0, \forall y \in \mathbb{R} : I \nsubseteq B(y, r)$$

Or equivalently (to bring about the sequential format, we use first countability):

$$\forall n \in \mathbb{N}, \forall y \in \mathbb{R}, \exists x \in I : x \notin B(y, n+1)$$

Choose some arbitrary $y \in \mathbb{R}$. Then $\forall n \in \mathbb{N}$ choose x_n such that $x_n \notin B(y, n+1)$, so that:

$$\forall n \in \mathbb{N} : |x_n - y| > n + 1 > 1/2$$

Now assume there existed some z such that some subsequence indexed by $\{n_k\}_{k\in\mathbb{N}}$ converged to z. Then:

$$\forall k \in \mathbb{N} : |x_{n_k} - y| \leqslant |x_{n_k} - z| + |y - z|$$

$$\Longrightarrow \forall k \in \mathbb{N} : ||x_{n_k} - y| - |y - z|| \leqslant |x_{n_k} - z|$$

Taking limit $k \to \infty$ we see that $n_k \to \infty$, thus:

$$||x_{n_k} - y| - |y - z|| \to 0$$

which is a contradiction because $|x_{n_k} - y| > n_k + 1$ which tells us that $|x_{n_k} - y| \to \infty$, so $x_{n_k} \nrightarrow z$ and since z is arbitrary, this holds true for all such $z \in \mathbb{R}$.

Lemma 1.3. *For any set $I \subseteq \mathbb{R}$, I is bounded if and only if it is totally bounded (ONLY for $\mathbb{R}$).*

Proof. $\Longleftarrow$: If I is bounded, then:

$$\exists r > 0, \exists y \in \mathbb{R} : I \subseteq B(y, r) \iff \exists r > 0, \exists y \in \mathbb{R}, \forall x \in I : x \in B(y, r)$$

We aim to show I is totally bounded, which means:

$$\forall \epsilon > 0, \exists x_0, ..., x_m \in I : I \subseteq \bigcup_{i=0}^{m} B(x_i, \epsilon)$$

This converse direction is necessarily true; however, the forward direction is not necessarily true in all spaces. Consider the following:

$$M := \max\{|x_i - x_j| : 0 \leqslant i, j \leqslant m\}$$

Then, choose some x_i; so without loss of generality, we choose x_0:

$$\forall x \in I, \exists x_i : x \in B(x_i, \epsilon) \implies |x - x_0| \leqslant |x - x_i| + |x_i - x_0| < \epsilon + M$$

$\Longrightarrow$: Now to prove the forward direction. We have that I is bounded, which means:

$$\exists y \in \mathbb{R}, \exists r > 0 : I \subseteq B(y, r)$$

Now we can show that $B(y, r)$ is totally bounded. Simply consider the uniform partition:

$$\Pi_B := \left\{ (y - r) + \frac{2r}{n} i : 0 \leqslant i \leqslant n \right\}$$

whereby:

$$\exists n \in \mathbb{N} : \frac{2r}{n} < \frac{\epsilon}{2}$$

$$\implies \forall \epsilon > 0, \exists \Pi_B : I \subseteq B(y, r) \subseteq \bigcup_{x_i \in \Pi_B} B(x_i, \epsilon/2)$$

Now $\forall x_i \in \Pi_B$ where $B(x_i, \epsilon/2) \cap I \neq \varnothing$, choose x_i' such that:

$$x_i' \in B(x_i, \epsilon/2) \cap I$$

and add it to the partition Π_I. Then we observe that:

$$\forall x \in I, \exists x_i \in \Pi_B : x \in B(x_i, \epsilon/2)$$

$$\implies \exists x_i' \in \Pi_I : |x - x_i'| \leqslant |x - x_i| + |x_i - x_i'| < \epsilon/2 + \epsilon/2 = \epsilon$$

thus, I is totally bounded such that,

$$\forall \epsilon > 0, \exists \Pi_I : I \subseteq \bigcup_{x_i' \in \Pi_I} B(x_i', \epsilon)$$

$\square$

Now the main assumption we made here is that every open ball in $\mathbb{R}$ is totally bounded (which we can show via uniform partitions). The rest of the argument is not unique to $\mathbb{R}$. This can give insight into what spaces obey this Heine-Borel compactness property.

1.2.1 Bounded/Totally-Bounded Equivalence

NOTE: I will look into this later, it has something to do with finite dimensions/functional analysis.

1.3 Limit Point Compactness

We now discuss an equivalent form of compactness called limit point compactness (which due to the first countability of $\mathbb{R}$) is essentially equivalent to the sequential characterization.

Definition 1.4. *A set $K \subseteq \mathbb{R}$ is limit point compact if for all infinite (countably or uncountably) sets $E \subseteq K$, there exists a limit point of E in K.*

Theorem 1.3. *A set K is limit point compact if and only if it is sequentially compact in $\mathbb{R}$.*

Proof. $\implies$: Now consider any sequence $\{x_n\}_{n\in\mathbb{N}} \in K^{\mathbb{N}}$, which is an infinite subset, so by limit point compactness it admits a limit point. Now:

$$\forall n \in \mathbb{N} : \{x_n\}_{n\in\mathbb{N}} \cap B\left(x, \frac{1}{n+1}\right) \neq \varnothing$$

So to construct the converging subsequence (note this requires the Axiom of Choice), we choose:

$$\forall k \in \mathbb{N} : x_{n_k} \in \{x_n\}_{n\in\mathbb{N}} \cap B\left(x, \frac{1}{k+1}\right)$$

such that $\forall k \in \mathbb{N} : n_{k+1} > n_k$. (**Not so sure about this proof because can we justify such a choice whereby $n_{k+1} > n_k$**).

An alternative prove is by contrapositive. Consider K is not sequentially compact, then:

$$\exists \{x_n\}_{n\in\mathbb{N}} \in K^{\mathbb{N}}, \forall \{n_k\}_{k\in\mathbb{N}}, \forall k \in \mathbb{N} : n_k < n_{k+1} \implies \forall x \in K : x_{n_k} \nrightarrow x$$

which means there exists a sequence with no convergent subsequence. In that case, the sequence is a subsequence of itself, which means there exists an infinite subset of K with no limit point, which follows from:

$$\exists x \in K : x \text{ is a limit point of } \{x_n\}_{n\in\mathbb{N}} \iff \exists \{n_k\}_{k\in\mathbb{N}} : n_k < n_{k+1} \wedge x_{n_k} \to x$$

the proof of which is provided in the section about nets.

$\impliedby$: Consider any infinite subset of K. Then taking a countable infinite subset of that, we obtain a sequence $\{x_n\}_{n\in\mathbb{N}} \in K^{\mathbb{N}}$ of DISTINCT points.

By sequential compactness, there exists a convergent subsequence converging to a point x. Notice that we do not want one value repeating infinitely many times; otherwise we may not satisfy sequential characterization of limit points.

Now it is possible that x appears as a value, but at most once within the subsequence. Ommission of x from this subsequence still results in a convergent subsequence. Assume x appears at some index n_j, then:

$$\forall \epsilon > 0, \exists k_0 \geqslant 0, \forall k \geqslant k_0 : |x_{n_k} - x| < \epsilon$$

so if $j \geqslant k_0$, we still have infinite values other than x from the subsequence within the ϵ-ball around x. This is trivial if $j < k_0$, thus x is an ω-limit point or equivalently a limit point. $\square$

Notice that the $\implies$ direction is not always true in general topological spaces as we used the first countability of metric spaces. The equivalence really stems from the idea of nets/countable nets or sequences. Limit point compactness is identical to sequential compactness in first countable spaces; the only difference being the former describes net-wise behavior, while the latter describes sequential.

But the equivalence fundamentally comes from the equivalent characterization of limit points.

2 Topological Compactness

3 Integration and Nets